HPC-VS Connectivity and Past-Year 5+ Drinking Frequency

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HPC-VS Binge Drinking Analysis

This R Markdown file runs the analyses and creates the figures reported in the manuscript and supplement:

1. Primary proportional-odds ordinal logistic models.
2. Permutation likelihood-ratio checks with 10,000 permutations.
3. DFBETAS influence sensitivity checks.
4. Sex-moderation models and conditional slopes.
5. Age sensitivity analyses.
6. Three-level alcohol-diagnosis follow-ups: no lifetime DSM-IV alcohol diagnosis, lifetime DSM-IV alcohol abuse, and lifetime DSM-IV alcohol dependence.
7. Diagnosis-group ANCOVAs for standardized left and right HPC-VS connectivity.
8. Manuscript Figures 2-5.

- `set.seed(23)`
- `N_PERM <- 10000`
- DFBETAS screens include max `|DFBETAS| > .20` and `2/sqrt(N)`

Statistical Models and Analyses

Main-Effect Models

Left Hemisphere

```r
left_base <- fit_polr_safe(y ~ c_ICV, data = left_dat)
left_main <- fit_polr_safe(y ~ c_ICV + c_left_hpc_vs, data = left_dat)

summary(left_main)
```

```
## Call:
## MASS::polr(formula = formula, data = data, Hess = TRUE, model = TRUE,
##     method = "logistic")
##
## Coefficients:
##                 Value Std. Error t value
## c_ICV          0.4209    0.07739   5.438
## c_left_hpc_vs 0.1571    0.07616   2.062
```

```
##
## Intercepts:
##                                Value   Std. Error t value
## Never|1-11 d/year              -0.6438  0.0877    -7.3398
## 1-11 d/year|1-3 d/month         0.5849  0.0870     6.7220
## 1-3 d/month|Weekly or greater  1.5059  0.1066    14.1310
##
## Residual Deviance: 1564.239
## AIC: 1574.239
```

```r
anova(left_base, left_main)
```

```
## Likelihood ratio tests of ordinal regression models
##
## Response: y
##                     Model Resid. df Resid. Dev   Test    Df LR stat.    Pr(Chi)
## 1                   c_ICV       591   1568.521
## 2 c_ICV + c_left_hpc_vs       590   1564.239 1 vs 2     1 4.281998 0.03851805
```

```r
left_main_lr <- lr_test(
  left_base,
  left_main,
  "Left main effect: add HPC-VS connectivity"
)
left_main_lr
```

```
## # A tibble: 1 x 4
##   test                                         df    LR      p
##   <chr>                                     <int> <dbl>  <dbl>
## 1 Left main effect: add HPC-VS connectivity     1  4.28 0.0385
```

```r
left_main_perm <- get_permutation_result(
  y ~ c_ICV,
  y ~ c_ICV + c_left_hpc_vs,
  left_dat,
  "Permutation LR, left main effect"
)
left_main_perm
```

```
## # A tibble: 1 x 4
##   test                                LR p_perm n_success
##   <chr>                            <dbl>  <dbl>     <int>
## 1 Permutation LR, left main effect  4.28  0.036     10000
```

```r
left_main_dfb <- run_dfbetas_checks(
  "left_main",
  left_dat,
  y ~ c_ICV,
  y ~ c_ICV + c_left_hpc_vs,
  "c_left_hpc_vs"
)
left_main_dfb$result
```

```
## # A tibble: 3 x 14
##   model cutoff_label  cutoff removed_n retained_n target_term     b     SE     z
##   <chr> <chr>          <dbl>     <int>      <int> <chr>       <dbl>  <dbl> <dbl>
## 1 left~ Full sample  NA              0        595 c_left_hpc~ 0.157 0.0762  2.06
## 2 left~ max |DFBETA~  0.2            5        590 c_left_hpc~ 0.186 0.0782  2.38
## 3 left~ max |DFBETA~  0.0820        59        536 c_left_hpc~ 0.233 0.0887  2.63
## # i 5 more variables: p <dbl>, OR <dbl>, LR <dbl>, LR_df <int>, LR_p <dbl>
```

Right Hemisphere

```r
right_base <- fit_polr_safe(y ~ c_ICV, data = right_dat)
right_main <- fit_polr_safe(y ~ c_ICV + c_right_hpc_vs, data = right_dat)

summary(right_main)
```

```
## Call:
## MASS::polr(formula = formula, data = data, Hess = TRUE, model = TRUE,
##     method = "logistic")
##
## Coefficients:
##                  Value Std. Error t value
## c_ICV          0.42935     0.0776  5.5326
## c_right_hpc_vs 0.01384     0.0747  0.1852
##
## Intercepts:
##                              Value   Std. Error t value
## Never|1-11 d/year            -0.6406  0.0875    -7.3198
## 1-11 d/year|1-3 d/month        0.5832  0.0868     6.7162
## 1-3 d/month|Weekly or greater  1.5004  0.1063    14.1146
##
## Residual Deviance: 1568.487
## AIC: 1578.487
```

```r
anova(right_base, right_main)
```

```
## Likelihood ratio tests of ordinal regression models
##
## Response: y
##                   Model Resid. df Resid. Dev   Test    Df   LR stat.  Pr(Chi)
## 1                 c_ICV       591   1568.521
## 2 c_ICV + c_right_hpc_vs       590   1568.487 1 vs 2     1 0.03433125 0.853004
```

```r
right_main_lr <- lr_test(
  right_base,
  right_main,
  "Right main effect: add HPC-VS connectivity"
)
right_main_lr
```

```
## # A tibble: 1 x 4
```

```
##   test                                          df     LR     p
##   <chr>                                      <int>  <dbl> <dbl>
## 1 Right main effect: add HPC-VS connectivity     1 0.0343 0.853
```

```
right_main_perm <- get_permutation_result(
  y ~ c_ICV,
  y ~ c_ICV + c_right_hpc_vs,
  right_dat,
  "Permutation LR, right main effect"
)
right_main_perm
```

```
## # A tibble: 1 x 4
##   test                                 LR p_perm n_success
##   <chr>                             <dbl>  <dbl>     <int>
## 1 Permutation LR, right main effect 0.034   0.85     10000
```

```
right_main_dfb <- run_dfbetas_checks(
  "right_main",
  right_dat,
  y ~ c_ICV,
  y ~ c_ICV + c_right_hpc_vs,
  "c_right_hpc_vs"
)
right_main_dfb$result
```

```
## # A tibble: 3 x 14
##   model    cutoff_label  cutoff removed_n retained_n target_term        b     SE
##   <chr>    <chr>          <dbl>     <int>      <int> <chr>          <dbl>  <dbl>
## 1 right_m~ Full sample   NA             0        595 c_right_hp~  1.38e-2 0.0747
## 2 right_m~ max |DFBETA~   0.2           4        591 c_right_hp~  2.70e-2 0.0759
## 3 right_m~ max |DFBETA~   0.0820       60        535 c_right_hp~ -2.97e-4 0.0878
## # i 6 more variables: z <dbl>, p <dbl>, OR <dbl>, LR <dbl>, LR_df <int>,
## #   LR_p <dbl>
```

Sex-Moderation Models

Left Hemisphere

```
left_sex <- fit_polr_safe(y ~ c_ICV + c_left_hpc_vs + Sex, data = left_dat)
left_int <- fit_polr_safe(y ~ c_ICV + c_left_hpc_vs * Sex, data = left_dat)

summary(left_int)
```

```
## Call:
## MASS::polr(formula = formula, data = data, Hess = TRUE, model = TRUE,
##     method = "logistic")
##
## Coefficients:
##                         Value Std. Error  t value
```

```
## c_ICV               0.190947    0.09799  1.94867
## c_left_hpc_vs       0.142264    0.07675  1.85354
## Sex1               -0.365977    0.09875 -3.70604
## c_left_hpc_vs:Sex1 -0.001648    0.07669 -0.02148
##
## Intercepts:
##                               Value   Std. Error t value
## Never|1-11 d/year             -0.7113  0.0905    -7.8586
## 1-11 d/year|1-3 d/month        0.5370  0.0887     6.0572
## 1-3 d/month|Weekly or greater  1.4790  0.1076    13.7491
##
## Residual Deviance: 1550.408
## AIC: 1564.408
```

```
anova(left_main, left_sex, left_int)
```

```
## Likelihood ratio tests of ordinal regression models
##
## Response: y
##                         Model Resid. df Resid. Dev   Test    Df     LR stat.
## 1       c_ICV + c_left_hpc_vs       590   1564.239
## 2 c_ICV + c_left_hpc_vs + Sex       589   1550.409 1 vs 2     1 1.383058e+01
## 3 c_ICV + c_left_hpc_vs * Sex       588   1550.408 2 vs 3     1 4.617184e-04
##        Pr(Chi)
## 1
## 2 0.0002000535
## 3 0.9828566685
```

```
left_sex_lr <- lr_test(left_main, left_sex, "Left model: add sex")
left_int_lr <- lr_test(
  left_sex,
  left_int,
  "Left model: add HPC-VS connectivity x Sex"
)
left_sex_lr
```

```
## # A tibble: 1 x 4
##   test                   df    LR        p
##   <chr>               <int> <dbl>    <dbl>
## 1 Left model: add sex     1  13.8 0.000200
```

```
left_int_lr
```

```
## # A tibble: 1 x 4
##   test                                         df       LR     p
##   <chr>                                     <int>    <dbl> <dbl>
## 1 Left model: add HPC-VS connectivity x Sex     1 0.000462 0.983
```

```
left_sex_perm <- get_permutation_result(
  y ~ c_ICV + c_left_hpc_vs,
  y ~ c_ICV + c_left_hpc_vs + Sex,
```

```r
  left_dat,
  "Permutation LR, left model add sex"
)
left_int_perm <- get_permutation_result(
  y ~ c_ICV + c_left_hpc_vs + Sex,
  y ~ c_ICV + c_left_hpc_vs * Sex,
  left_dat,
  "Permutation LR, left interaction"
)
left_sex_perm
```

```
## # A tibble: 1 x 4
##   test                                  LR    p_perm n_success
##   <chr>                              <dbl>     <dbl>     <int>
## 1 Permutation LR, left model add sex  13.8 0.0001000     10000
```

```r
left_int_perm
```

```
## # A tibble: 1 x 4
##   test                                LR p_perm n_success
##   <chr>                            <dbl>  <dbl>     <int>
## 1 Permutation LR, left interaction     0  0.985     10000
```

```r
left_int_dfb <- run_dfbetas_checks(
  "left_interaction",
  left_dat,
  y ~ c_ICV + c_left_hpc_vs + Sex,
  y ~ c_ICV + c_left_hpc_vs * Sex,
  "c_left_hpc_vs:Sex1"
)
left_int_dfb$result
```

```
## # A tibble: 3 x 14
##   model    cutoff_label  cutoff removed_n retained_n target_term         b     SE
##   <chr>    <chr>          <dbl>     <int>      <int> <chr>           <dbl>  <dbl>
## 1 left_in~ Full sample  NA              0        595 c_left_hpc~ -1.65e-3 0.0767
## 2 left_in~ max |DFBETA~  0.2            4        591 c_left_hpc~  4.82e-4 0.0783
## 3 left_in~ max |DFBETA~  0.0820       100        495 c_left_hpc~  8.97e-3 0.0934
## # i 6 more variables: z <dbl>, p <dbl>, OR <dbl>, LR <dbl>, LR_df <int>,
## #   LR_p <dbl>
```

```r
emmeans::emtrends(
  left_int,
  specs = ~ Sex,
  var = "c_left_hpc_vs",
  mode = "latent"
)
```

```
##  Sex c_left_hpc_vs.trend    SE  df asymp.LCL asymp.UCL
##  F                 0.141 0.103 Inf   -0.0610     0.342
##  M                 0.144 0.114 Inf   -0.0793     0.367
##
## Confidence level used: 0.95
```

```r
left_slopes_full <- simple_slopes(left_int, "c_left_hpc_vs", "Full sample")
left_slopes_rms <- simple_slopes(
  left_int_dfb$rms$filtered_model,
  "c_left_hpc_vs",
  "After max |DFBETAS| > .20 removal"
)
left_slopes_bkw <- simple_slopes(
  left_int_dfb$bkw$filtered_model,
  "c_left_hpc_vs",
  "After max |DFBETAS| > 2/sqrt(N) removal"
)
left_slopes_full
```

```
## # A tibble: 3 x 8
##   analysis    contrast        b    SE       z     p lower.CL upper.CL
##   <chr>       <chr>       <dbl> <dbl>   <dbl> <dbl>    <dbl>    <dbl>
## 1 Full sample F         0.141   0.103  1.37   0.172  -0.0610    0.342
## 2 Full sample M         0.144   0.114  1.26   0.206  -0.0793    0.367
## 3 Full sample F - M    -0.00330 0.153 -0.0215 0.983  -0.304     0.297
```

```r
left_slopes_rms
```

```
## # A tibble: 3 x 8
##   analysis                contrast       b    SE       z     p lower.CL upper.CL
##   <chr>                   <chr>      <dbl> <dbl>   <dbl> <dbl>    <dbl>    <dbl>
## 1 After max |DFBETAS| > ~ F        1.69e-1 0.105 1.61    0.108  -0.0373    0.376
## 2 After max |DFBETAS| > ~ M        1.68e-1 0.116 1.45    0.147  -0.0591    0.396
## 3 After max |DFBETAS| > ~ F - M    9.63e-4 0.157 0.00615 0.995  -0.306     0.308
```

```r
left_slopes_bkw
```

```
## # A tibble: 3 x 8
##   analysis                 contrast      b    SE      z      p lower.CL upper.CL
##   <chr>                    <chr>     <dbl> <dbl>  <dbl>  <dbl>    <dbl>    <dbl>
## 1 After max |DFBETAS| > 2~ F        0.219  0.119 1.84   0.0654  -0.0140    0.453
## 2 After max |DFBETAS| > 2~ M        0.201  0.144 1.40   0.162   -0.0806    0.483
## 3 After max |DFBETAS| > 2~ F - M    0.0179 0.187 0.0960 0.923   -0.348     0.384
```

Right Hemisphere

```r
right_sex <- fit_polr_safe(y ~ c_ICV + c_right_hpc_vs + Sex, data = right_dat)
right_int <- fit_polr_safe(y ~ c_ICV + c_right_hpc_vs * Sex, data = right_dat)

summary(right_int)
```

```
## Call:
## MASS::polr(formula = formula, data = data, Hess = TRUE, model = TRUE,
##     method = "logistic")
##
```

```
## Coefficients:
##                         Value Std. Error t value
## c_ICV                 0.19331    0.09854  1.9617
## c_right_hpc_vs       -0.01576    0.07724 -0.2041
## Sex1                 -0.38235    0.09900 -3.8622
## c_right_hpc_vs:Sex1  0.13306    0.07721  1.7234
##
## Intercepts:
##                                Value   Std. Error t value
## Never|1-11 d/year              -0.7249  0.0908    -7.9860
## 1-11 d/year|1-3 d/month         0.5231  0.0886     5.9010
## 1-3 d/month|Weekly or greater  1.4664  0.1072    13.6824
##
## Residual Deviance: 1550.883
## AIC: 1564.883
```

```r
anova(right_main, right_sex, right_int)
```

```
## Likelihood ratio tests of ordinal regression models
##
## Response: y
##                           Model Resid. df Resid. Dev   Test    Df  LR stat.
## 1       c_ICV + c_right_hpc_vs       590   1568.487
## 2 c_ICV + c_right_hpc_vs + Sex       589   1553.883 1 vs 2     1 14.603517
## 3 c_ICV + c_right_hpc_vs * Sex       588   1550.883 2 vs 3     1  3.000699
##        Pr(Chi)
## 1
## 2 0.0001326667
## 3 0.0832285929
```

```r
right_sex_lr <- lr_test(right_main, right_sex, "Right model: add sex")
right_int_lr <- lr_test(
  right_sex,
  right_int,
  "Right model: add HPC-VS connectivity x Sex"
)
right_sex_lr
```

```
## # A tibble: 1 x 4
##   test                    df    LR        p
##   <chr>                <int> <dbl>    <dbl>
## 1 Right model: add sex     1  14.6 0.000133
```

```r
right_int_lr
```

```
## # A tibble: 1 x 4
##   test                                          df    LR      p
##   <chr>                                      <int> <dbl>  <dbl>
## 1 Right model: add HPC-VS connectivity x Sex     1  3.00 0.0832
```

```r
right_sex_perm <- get_permutation_result(
  y ~ c_ICV + c_right_hpc_vs,
  y ~ c_ICV + c_right_hpc_vs + Sex,
  right_dat,
  "Permutation LR, right model add sex"
)
right_int_perm <- get_permutation_result(
  y ~ c_ICV + c_right_hpc_vs + Sex,
  y ~ c_ICV + c_right_hpc_vs * Sex,
  right_dat,
  "Permutation LR, right interaction"
)
right_sex_perm
```

```
## # A tibble: 1 x 4
##   test                                   LR    p_perm n_success
##   <chr>                               <dbl>     <dbl>     <int>
## 1 Permutation LR, right model add sex  14.6 0.0001000     10000
```

```r
right_int_perm
```

```
## # A tibble: 1 x 4
##   test                                 LR p_perm n_success
##   <chr>                             <dbl>  <dbl>     <int>
## 1 Permutation LR, right interaction  3.00  0.082     10000
```

```r
right_int_dfb <- run_dfbetas_checks(
  "right_interaction",
  right_dat,
  y ~ c_ICV + c_right_hpc_vs + Sex,
  y ~ c_ICV + c_right_hpc_vs * Sex,
  "c_right_hpc_vs:Sex1"
)
right_int_dfb$result
```

```
## # A tibble: 3 x 14
##   model cutoff_label  cutoff removed_n retained_n target_term     b     SE     z
##   <chr> <chr>          <dbl>     <int>      <int> <chr>       <dbl>  <dbl> <dbl>
## 1 righ~ Full sample   NA             0        595 c_right_hp~ 0.133 0.0772  1.72
## 2 righ~ max |DFBETA~   0.2           5        590 c_right_hp~ 0.171 0.0830  2.07
## 3 righ~ max |DFBETA~   0.0820      101        494 c_right_hp~ 0.252 0.0967  2.60
## # i 5 more variables: p <dbl>, OR <dbl>, LR <dbl>, LR_df <int>, LR_p <dbl>
```

```r
emmeans::emtrends(
  right_int,
  specs = ~ Sex,
  var = "c_right_hpc_vs",
  mode = "latent"
)
```

```
##  Sex c_right_hpc_vs.trend     SE  df asymp.LCL asymp.UCL
```

```
##  F                  0.117 0.0999 Inf   -0.0785     0.313
##  M                 -0.149 0.1180 Inf   -0.3797     0.082
##
## Confidence level used: 0.95
```

```r
right_slopes_full <- simple_slopes(right_int, "c_right_hpc_vs", "Full sample")
right_slopes_rms <- simple_slopes(
  right_int_dfb$rms$filtered_model,
  "c_right_hpc_vs",
  "After max |DFBETAS| > .20 removal"
)
right_slopes_bkw <- simple_slopes(
  right_int_dfb$bkw$filtered_model,
  "c_right_hpc_vs",
  "After max |DFBETAS| > 2/sqrt(N) removal"
)
right_slopes_full
```

```
## # A tibble: 3 x 8
##   analysis    contrast      b     SE     z      p lower.CL upper.CL
##   <chr>       <chr>     <dbl>  <dbl> <dbl>  <dbl>    <dbl>    <dbl>
## 1 Full sample F         0.117 0.0999  1.17 0.240   -0.0785   0.313
## 2 Full sample M        -0.149 0.118  -1.26 0.206   -0.380    0.0820
## 3 Full sample F - M     0.266 0.154   1.72 0.0848  -0.0365   0.569
```

```r
right_slopes_rms
```

```
## # A tibble: 3 x 8
##   analysis                   contrast      b    SE     z      p lower.CL upper.CL
##   <chr>                      <chr>     <dbl> <dbl> <dbl>  <dbl>    <dbl>    <dbl>
## 1 After max |DFBETAS| > .2~ F         0.163 0.103  1.57 0.116   -0.0399   0.365
## 2 After max |DFBETAS| > .2~ M        -0.180 0.130 -1.39 0.165   -0.435    0.0745
## 3 After max |DFBETAS| > .2~ F - M     0.343 0.166  2.07 0.0389   0.0175   0.668
```

```r
right_slopes_bkw
```

```
## # A tibble: 3 x 8
##   analysis                  contrast      b    SE     z       p lower.CL upper.CL
##   <chr>                     <chr>     <dbl> <dbl> <dbl>   <dbl>    <dbl>    <dbl>
## 1 After max |DFBETAS| > 2~ F         0.193 0.115  1.67 0.0948   -0.0334  0.419
## 2 After max |DFBETAS| > 2~ M        -0.311 0.155 -2.01 0.0450   -0.614  -0.00698
## 3 After max |DFBETAS| > 2~ F - M     0.503 0.193  2.60 0.00926   0.124   0.882
```

Age Sensitivity Models

```r
left_age_reduced <- fit_polr_safe(y ~ c_ICV + c_Age, data = left_dat)
left_age_full <- fit_polr_safe(
  y ~ c_ICV + c_Age + c_left_hpc_vs,
  data = left_dat
)
summary(left_age_full)
```

```
## Call:
## MASS::polr(formula = formula, data = data, Hess = TRUE, model = TRUE,
##     method = "logistic")
##
## Coefficients:
##                 Value Std. Error t value
## c_ICV          0.3897    0.07828   4.979
## c_Age         -0.1887    0.07685  -2.456
## c_left_hpc_vs  0.1445    0.07636   1.892
##
## Intercepts:
##                              Value   Std. Error t value
## Never|1-11 d/year            -0.6511  0.0881    -7.3909
## 1-11 d/year|1-3 d/month       0.5868  0.0873     6.7243
## 1-3 d/month|Weekly or greater  1.5143  0.1069    14.1684
##
## Residual Deviance: 1558.182
## AIC: 1570.182
```

```r
anova(left_age_reduced, left_age_full)
```

```
## Likelihood ratio tests of ordinal regression models
##
## Response: y
##                           Model Resid. df Resid. Dev   Test    Df LR stat.
## 1                 c_ICV + c_Age       590   1561.783
## 2 c_ICV + c_Age + c_left_hpc_vs       589   1558.182 1 vs 2     1 3.601038
##      Pr(Chi)
## 1
## 2 0.05774349
```

```r
left_age_lr <- lr_test(
  left_age_reduced,
  left_age_full,
  "Left + age: add HPC-VS connectivity"
)

right_age_reduced <- fit_polr_safe(
  y ~ c_ICV + c_Age + c_right_hpc_vs + Sex,
  data = right_dat
)
right_age_full <- fit_polr_safe(
  y ~ c_ICV + c_Age + c_right_hpc_vs * Sex,
  data = right_dat
)
summary(right_age_full)
```

```
## Call:
## MASS::polr(formula = formula, data = data, Hess = TRUE, model = TRUE,
##     method = "logistic")
##
## Coefficients:
##                         Value Std. Error t value
```

```
## c_ICV                0.17580    0.09879  1.7795
## c_Age               -0.17724    0.07741 -2.2896
## c_right_hpc_vs      -0.02208    0.07746 -0.2851
## Sex1                -0.36275    0.09951 -3.6453
## c_right_hpc_vs:Sex1  0.13735    0.07742  1.7741
##
## Intercepts:
##                              Value   Std. Error t value
## Never|1-11 d/year            -0.7296  0.0912    -8.0026
## 1-11 d/year|1-3 d/month       0.5264  0.0889     5.9196
## 1-3 d/month|Weekly or greater  1.4751  0.1075    13.7249
##
## Residual Deviance: 1545.619
## AIC: 1561.619
```

```
anova(right_age_reduced, right_age_full)
```

```
## Likelihood ratio tests of ordinal regression models
##
## Response: y
##                                   Model Resid. df Resid. Dev   Test    Df
## 1 c_ICV + c_Age + c_right_hpc_vs + Sex       588   1548.801
## 2 c_ICV + c_Age + c_right_hpc_vs * Sex       587   1545.619 1 vs 2     1
##   LR stat.    Pr(Chi)
## 1
## 2 3.182248 0.07444223
```

```
right_age_lr <- lr_test(
  right_age_reduced,
  right_age_full,
  "Right + age: add HPC-VS connectivity x Sex"
)

left_age_mod_reduced <- fit_polr_safe(
  y ~ c_ICV + c_Age + c_left_hpc_vs,
  data = left_dat
)
left_age_mod_full <- fit_polr_safe(
  y ~ c_ICV + c_Age * c_left_hpc_vs,
  data = left_dat
)
summary(left_age_mod_full)
```

```
## Call:
## MASS::polr(formula = formula, data = data, Hess = TRUE, model = TRUE,
##     method = "logistic")
##
## Coefficients:
##                        Value Std. Error t value
## c_ICV                0.39005    0.07832  4.9803
## c_Age               -0.19119    0.07708 -2.4803
## c_left_hpc_vs        0.14897    0.07704  1.9336
## c_Age:c_left_hpc_vs  0.03286    0.08145  0.4034
```

```
##
## Intercepts:
##                               Value   Std. Error t value
## Never|1-11 d/year             -0.6538  0.0884    -7.3986
## 1-11 d/year|1-3 d/month        0.5847  0.0874     6.6872
## 1-3 d/month|Weekly or greater  1.5123  0.1070    14.1351
##
## Residual Deviance: 1558.019
## AIC: 1572.019
```

```
anova(left_age_mod_reduced, left_age_mod_full)
```

```
## Likelihood ratio tests of ordinal regression models
##
## Response: y
##                             Model Resid. df Resid. Dev   Test    Df  LR stat.
## 1 c_ICV + c_Age + c_left_hpc_vs       589   1558.182
## 2 c_ICV + c_Age * c_left_hpc_vs       588   1558.019 1 vs 2     1 0.1624927
##     Pr(Chi)
## 1
## 2 0.6868719
```

```
left_age_mod_lr <- lr_test(
  left_age_mod_reduced,
  left_age_mod_full,
  "Left: add HPC-VS connectivity x Age"
)

right_age_mod_reduced <- fit_polr_safe(
  y ~ c_ICV + c_Age + c_right_hpc_vs * Sex,
  data = right_dat
)
right_age_mod_full <- fit_polr_safe(
  y ~ c_ICV + c_Age + c_right_hpc_vs * Sex +
    c_right_hpc_vs:c_Age + Sex:c_Age +
    c_right_hpc_vs:Sex:c_Age,
  data = right_dat
)
summary(right_age_mod_full)
```

```
## Call:
## MASS::polr(formula = formula, data = data, Hess = TRUE, model = TRUE,
##     method = "logistic")
##
## Coefficients:
##                                Value Std. Error   t value
## c_ICV                      0.1586175    0.09946  1.594843
## c_Age                     -0.1684954    0.07825 -2.153261
## c_right_hpc_vs            -0.0067636    0.07880 -0.085828
## Sex1                      -0.3834475    0.10011 -3.830266
## c_right_hpc_vs:Sex1        0.1114718    0.07871  1.416196
## c_Age:c_right_hpc_vs       0.1382672    0.08475  1.631432
## c_Age:Sex1                -0.1185479    0.07839 -1.512376
```

```
## c_Age:c_right_hpc_vs:Sex1  0.0003083    0.08477  0.003636
##
## Intercepts:
##                               Value   Std. Error t value
## Never|1-11 d/year             -0.7623  0.0932    -8.1819
## 1-11 d/year|1-3 d/month        0.5032  0.0901     5.5855
## 1-3 d/month|Weekly or greater  1.4534  0.1083    13.4228
##
## Residual Deviance: 1540.091
## AIC: 1562.091
```

```r
anova(right_age_mod_reduced, right_age_mod_full)
```

```
## Likelihood ratio tests of ordinal regression models
##
## Response: y
##                                                                                                Model
## 1                                                                c_ICV + c_Age + c_right_hpc_vs * Sex
## 2 c_ICV + c_Age + c_right_hpc_vs * Sex + c_right_hpc_vs:c_Age + Sex:c_Age + c_right_hpc_vs:Sex:c_Age
##   Resid. df Resid. Dev   Test    Df LR stat.   Pr(Chi)
## 1       587   1545.619
## 2       584   1540.091 1 vs 2     3 5.527608 0.1369967
```

```r
right_age_mod_lr <- lr_test(
  right_age_mod_reduced,
  right_age_mod_full,
  "Right: add age moderation terms"
)

left_age_lr
```

```
## # A tibble: 1 x 4
##   test                                  df    LR      p
##   <chr>                              <int> <dbl>  <dbl>
## 1 Left + age: add HPC-VS connectivity     1  3.60 0.0577
```

```r
right_age_lr
```

```
## # A tibble: 1 x 4
##   test                                        df    LR      p
##   <chr>                                    <int> <dbl>  <dbl>
## 1 Right + age: add HPC-VS connectivity x Sex     1  3.18 0.0744
```

```r
left_age_mod_lr
```

```
## # A tibble: 1 x 4
##   test                                  df    LR     p
##   <chr>                              <int> <dbl> <dbl>
## 1 Left: add HPC-VS connectivity x Age     1 0.162 0.687
```

```r
right_age_mod_lr
```

```
## # A tibble: 1 x 4
##   test                               df    LR     p
##   <chr>                           <int> <dbl> <dbl>
## 1 Right: add age moderation terms     3  5.53 0.137
```

Alcohol Diagnosis Analyses

```r
sex_totals <- table(analysis_dat$Sex)
total_n <- nrow(analysis_dat)
female_n <- unname(sex_totals["F"])
male_n <- unname(sex_totals["M"])
female_abuse <- sum(analysis_dat$Sex == "F" & analysis_dat$alcohol_abuse_dx)
male_abuse <- sum(analysis_dat$Sex == "M" & analysis_dat$alcohol_abuse_dx)
total_abuse <- sum(analysis_dat$alcohol_abuse_dx)
female_dependence <- sum(analysis_dat$Sex == "F" & analysis_dat$alcohol_dependence_dx)
male_dependence <- sum(analysis_dat$Sex == "M" & analysis_dat$alcohol_dependence_dx)
total_dependence <- sum(analysis_dat$alcohol_dependence_dx)
female_aud <- sum(analysis_dat$Sex == "F" & analysis_dat$AUD == "AUD")
male_aud <- sum(analysis_dat$Sex == "M" & analysis_dat$AUD == "AUD")
total_aud <- sum(analysis_dat$AUD == "AUD")

diagnosis_counts <- tibble(
  characteristic = c(
    "Sample size",
    "Alcohol abuse",
    "Alcohol dependence",
    "Any alcohol use disorder"
  ),
  female_n = c(female_n, female_abuse, female_dependence, female_aud),
  male_n = c(male_n, male_abuse, male_dependence, male_aud),
  total_n = c(total_n, total_abuse, total_dependence, total_aud)
)

diagnosis_counts
```

```
## # A tibble: 4 x 4
##   characteristic           female_n male_n total_n
##   <chr>                       <int>  <int>   <int>
## 1 Sample size                   333    262     595
## 2 Alcohol abuse                  45     60     105
## 3 Alcohol dependence              9     15      24
## 4 Any alcohol use disorder       54     75     129
```

```r
table(analysis_dat$DxGroup)
```

```
##
##     No AUD      Abuse Dependence
##        466        105         24
```

```r
table(analysis_dat$Sex, analysis_dat$DxGroup)
```

```
##
##     No AUD Abuse Dependence
##   F    279    45          9
##   M    187    60         15
```

```r
left_dx_mod_reduced <- fit_polr_safe(
  y ~ c_ICV + DxGroup + c_left_hpc_vs,
  data = analysis_dat
)
left_dx_mod_full <- fit_polr_safe(
  y ~ c_ICV + DxGroup * c_left_hpc_vs,
  data = analysis_dat
)
summary(left_dx_mod_full)
```

```
## Call:
## MASS::polr(formula = formula, data = data, Hess = TRUE, model = TRUE,
##     method = "logistic")
##
## Coefficients:
##                                     Value Std. Error t value
## c_ICV                            0.389846    0.07852 4.96494
## DxGroupAbuse                     1.412945    0.20728 6.81661
## DxGroupDependence                1.930130    0.43117 4.47649
## c_left_hpc_vs                    0.135004    0.08850 1.52540
## DxGroupAbuse:c_left_hpc_vs       0.006344    0.19007 0.03338
## DxGroupDependence:c_left_hpc_vs 0.225149    0.43555 0.51693
##
## Intercepts:
##                               Value   Std. Error t value
## Never|1-11 d/year             -0.3989  0.0942    -4.2358
## 1-11 d/year|1-3 d/month        0.9156  0.1002     9.1376
## 1-3 d/month|Weekly or greater  1.9327  0.1262    15.3171
##
## Residual Deviance: 1499.224
## AIC: 1517.224
```

```r
anova(left_dx_mod_reduced, left_dx_mod_full)
```

```
## Likelihood ratio tests of ordinal regression models
##
## Response: y
##                               Model Resid. df Resid. Dev   Test    Df  LR stat.
## 1 c_ICV + DxGroup + c_left_hpc_vs       588   1499.493
## 2 c_ICV + DxGroup * c_left_hpc_vs       586   1499.224 1 vs 2     2 0.2685635
##     Pr(Chi)
## 1
## 2 0.8743437
```

```r
left_dx_mod_lr <- lr_test(
  left_dx_mod_reduced,
  left_dx_mod_full,
  "Left: add diagnosis group x HPC-VS connectivity"
)

right_dx_mod_reduced <- fit_polr_safe(
  y ~ c_ICV + DxGroup + c_right_hpc_vs * Sex,
  data = analysis_dat
)
right_dx_mod_full <- fit_polr_safe(
  y ~ c_ICV + c_right_hpc_vs * Sex * DxGroup,
  data = analysis_dat
)
summary(right_dx_mod_full)
```

```
## Call:
## MASS::polr(formula = formula, data = data, Hess = TRUE, model = TRUE,
##     method = "logistic")
##
## Coefficients:
##                                          Value Std. Error  t value
## c_ICV                                 0.224554    0.10060  2.23220
## c_right_hpc_vs                       -0.009618    0.09103 -0.10566
## Sex1                                 -0.283995    0.10852 -2.61691
## DxGroupAbuse                          1.384319    0.20987  6.59597
## DxGroupDependence                     1.797637    0.45888  3.91748
## c_right_hpc_vs:Sex1                   0.058693    0.09094  0.64541
## c_right_hpc_vs:DxGroupAbuse           0.268106    0.19961  1.34316
## c_right_hpc_vs:DxGroupDependence     -0.540240    0.40165 -1.34505
## Sex1:DxGroupAbuse                     0.003005    0.20512  0.01465
## Sex1:DxGroupDependence               -0.698914    0.45488 -1.53648
## c_right_hpc_vs:Sex1:DxGroupAbuse      0.046870    0.19914  0.23536
## c_right_hpc_vs:Sex1:DxGroupDependence  0.551426    0.40074  1.37602
##
## Intercepts:
##                               Value   Std. Error t value
## Never|1-11 d/year             -0.4693  0.0985    -4.7633
## 1-11 d/year|1-3 d/month        0.8605  0.1030     8.3573
## 1-3 d/month|Weekly or greater  1.9080  0.1287    14.8199
##
## Residual Deviance: 1485.419
## AIC: 1515.419
```

```r
anova(right_dx_mod_reduced, right_dx_mod_full)
```

```
## Likelihood ratio tests of ordinal regression models
##
## Response: y
##                                      Model Resid. df Resid. Dev   Test    Df
## 1 c_ICV + DxGroup + c_right_hpc_vs * Sex       586   1492.682
## 2 c_ICV + c_right_hpc_vs * Sex * DxGroup       580   1485.419 1 vs 2     6
```

```
##   LR stat.  Pr(Chi)
## 1
## 2 7.262903 0.297217
```

```
right_dx_mod_lr <- lr_test(
  right_dx_mod_reduced,
  right_dx_mod_full,
  "Right: add diagnosis-group moderation terms"
)

left_dx_mod_lr
```

```
## # A tibble: 1 x 4
##   test                                               df    LR     p
##   <chr>                                           <int> <dbl> <dbl>
## 1 Left: add diagnosis group x HPC-VS connectivity     2 0.269 0.874
```

```
right_dx_mod_lr
```

```
## # A tibble: 1 x 4
##   test                                           df    LR     p
##   <chr>                                       <int> <dbl> <dbl>
## 1 Right: add diagnosis-group moderation terms     6  7.26 0.297
```

```
left_conn_reduced <- lm(c_left_hpc_vs ~ Sex + c_ICV, data = analysis_dat)
left_conn_full <- lm(
  c_left_hpc_vs ~ DxGroup + Sex + c_ICV,
  data = analysis_dat
)
right_conn_reduced <- lm(c_right_hpc_vs ~ Sex + c_ICV, data = analysis_dat)
right_conn_full <- lm(
  c_right_hpc_vs ~ DxGroup + Sex + c_ICV,
  data = analysis_dat
)

summary(left_conn_full)
```

```
##
## Call:
## lm(formula = c_left_hpc_vs ~ DxGroup + Sex + c_ICV, data = analysis_dat)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -4.2402 -0.7046 -0.0353  0.6447  3.7753
##
## Coefficients:
##                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)       -0.009048   0.047042  -0.192    0.848
## DxGroupAbuse       0.087012   0.108762   0.800    0.424
## DxGroupDependence  0.081837   0.210015   0.390    0.697
## Sex1              -0.080516   0.052887  -1.522    0.128
## c_ICV              0.030691   0.052332   0.586    0.558
```

```
##
## Residual standard error: 0.997 on 590 degrees of freedom
## Multiple R-squared:  0.01273,    Adjusted R-squared:  0.006033
## F-statistic: 1.901 on 4 and 590 DF,  p-value: 0.1087
```

```r
anova(left_conn_reduced, left_conn_full)
```

```
## Analysis of Variance Table
##
## Model 1: c_left_hpc_vs ~ Sex + c_ICV
## Model 2: c_left_hpc_vs ~ DxGroup + Sex + c_ICV
##   Res.Df    RSS Df Sum of Sq      F Pr(>F)
## 1    592 587.17
## 2    590 586.44  2   0.73174 0.3681 0.6922
```

```r
summary(right_conn_full)
```

```
##
## Call:
## lm(formula = c_right_hpc_vs ~ DxGroup + Sex + c_ICV, data = analysis_dat)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -4.9948 -0.5987 -0.0304  0.6246  4.2322
##
## Coefficients:
##                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)        0.017371   0.047071   0.369   0.7122
## DxGroupAbuse      -0.086802   0.108828  -0.798   0.4254
## DxGroupDependence -0.009537   0.210143  -0.045   0.9638
## Sex1              -0.013982   0.052919  -0.264   0.7917
## c_ICV              0.097257   0.052364   1.857   0.0638 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9976 on 590 degrees of freedom
## Multiple R-squared:  0.01153,    Adjusted R-squared:  0.004827
## F-statistic:  1.72 on 4 and 590 DF,  p-value: 0.1439
```

```r
anova(right_conn_reduced, right_conn_full)
```

```
## Analysis of Variance Table
##
## Model 1: c_right_hpc_vs ~ Sex + c_ICV
## Model 2: c_right_hpc_vs ~ DxGroup + Sex + c_ICV
##   Res.Df    RSS Df Sum of Sq      F Pr(>F)
## 1    592 587.79
## 2    590 587.15  2   0.63438 0.3187 0.7272
```

```r
left_conn_ancova <- lm_block_test(
  left_conn_reduced,
```

```r
  left_conn_full,
  "Left HPC-VS connectivity: add diagnosis group"
)
right_conn_ancova <- lm_block_test(
  right_conn_reduced,
  right_conn_full,
  "Right HPC-VS connectivity: add diagnosis group"
)

left_conn_emm <- emmeans::emmeans(left_conn_full, ~ DxGroup)
right_conn_emm <- emmeans::emmeans(right_conn_full, ~ DxGroup)

left_conn_emm
```

```
##  DxGroup      emmean     SE  df lower.CL upper.CL
##  No AUD     -0.00905 0.0470 590   -0.101   0.0833
##  Abuse       0.07796 0.0979 590   -0.114   0.2702
##  Dependence  0.07279 0.2040 590   -0.328   0.4736
##
## Results are averaged over the levels of: Sex
## Confidence level used: 0.95
```

```r
right_conn_emm
```

```
##  DxGroup      emmean     SE  df lower.CL upper.CL
##  No AUD      0.01737 0.0471 590  -0.0751    0.110
##  Abuse      -0.06943 0.0979 590  -0.2618    0.123
##  Dependence  0.00783 0.2040 590  -0.3932    0.409
##
## Results are averaged over the levels of: Sex
## Confidence level used: 0.95
```

```r
pairs(left_conn_emm)
```

```
##  contrast             estimate    SE  df t.ratio p.value
##  No AUD - Abuse       -0.08701 0.109 590  -0.800  0.7031
##  No AUD - Dependence -0.08184 0.210 590  -0.390  0.9197
##  Abuse - Dependence   0.00518 0.226 590   0.023  0.9997
##
## Results are averaged over the levels of: Sex
## P value adjustment: tukey method for comparing a family of 3 estimates
```

```r
pairs(right_conn_emm)
```

```
##  contrast             estimate    SE  df t.ratio p.value
##  No AUD - Abuse        0.08680 0.109 590   0.798  0.7046
##  No AUD - Dependence   0.00954 0.210 590   0.045  0.9989
##  Abuse - Dependence  -0.07726 0.227 590  -0.341  0.9379
##
## Results are averaged over the levels of: Sex
## P value adjustment: tukey method for comparing a family of 3 estimates
```

```r
diagnosis_connectivity_emmeans <- bind_rows(
  as_tibble(as.data.frame(summary(left_conn_emm))) %>%
    mutate(hemisphere = "Left", .before = 1),
  as_tibble(as.data.frame(summary(right_conn_emm))) %>%
    mutate(hemisphere = "Right", .before = 1)
)
diagnosis_connectivity_pairwise <- bind_rows(
  as_tibble(as.data.frame(summary(pairs(left_conn_emm), infer = c(TRUE, TRUE)))) %>%
    mutate(hemisphere = "Left", .before = 1),
  as_tibble(as.data.frame(summary(pairs(right_conn_emm), infer = c(TRUE, TRUE)))) %>%
    mutate(hemisphere = "Right", .before = 1)
)

diagnosis_connectivity_emmeans
```

```
## # A tibble: 6 x 7
##   hemisphere DxGroup      emmean     SE    df lower.CL upper.CL
##   <chr>      <fct>         <dbl>  <dbl> <dbl>    <dbl>    <dbl>
## 1 Left       No AUD     -0.00905 0.0470   590  -0.101    0.0833
## 2 Left       Abuse       0.0780  0.0979   590  -0.114    0.270
## 3 Left       Dependence  0.0728  0.204    590  -0.328    0.474
## 4 Right      No AUD      0.0174  0.0471   590  -0.0751   0.110
## 5 Right      Abuse      -0.0694  0.0979   590  -0.262    0.123
## 6 Right      Dependence  0.00783 0.204    590  -0.393    0.409
```

```r
diagnosis_connectivity_pairwise
```

```
## # A tibble: 6 x 9
##   hemisphere contrast     estimate    SE    df lower.CL upper.CL t.ratio p.value
##   <chr>      <chr>           <dbl> <dbl> <dbl>    <dbl>    <dbl>   <dbl>   <dbl>
## 1 Left       No AUD - Ab~ -0.0870  0.109   590   -0.343    0.169 -0.800    0.703
## 2 Left       No AUD - De~ -0.0818  0.210   590   -0.575    0.412 -0.390    0.920
## 3 Left       Abuse - Dep~  0.00518 0.226   590   -0.527    0.537  0.0229   1.000
## 4 Right      No AUD - Ab~  0.0868  0.109   590   -0.169    0.343  0.798    0.705
## 5 Right      No AUD - De~  0.00954 0.210   590   -0.484    0.503  0.0454   0.999
## 6 Right      Abuse - Dep~ -0.0773  0.227   590   -0.610    0.455 -0.341    0.938
```

Final Figure 2

This section displays the sex-stratified drinking-frequency distribution figure directly in the report.

```
## # A tibble: 8 x 4
##   Sex   y                      n  prop
##   <fct> <ord>              <int> <dbl>
## 1 F     Never                143 0.429
## 2 F     1-11 d/year          106 0.318
## 3 F     1-3 d/month           47 0.141
## 4 F     Weekly or greater     37 0.111
## 5 M     Never                 66 0.252
## 6 M     1-11 d/year           63 0.240
## 7 M     1-3 d/month           56 0.214
## 8 M     Weekly or greater     77 0.294
```

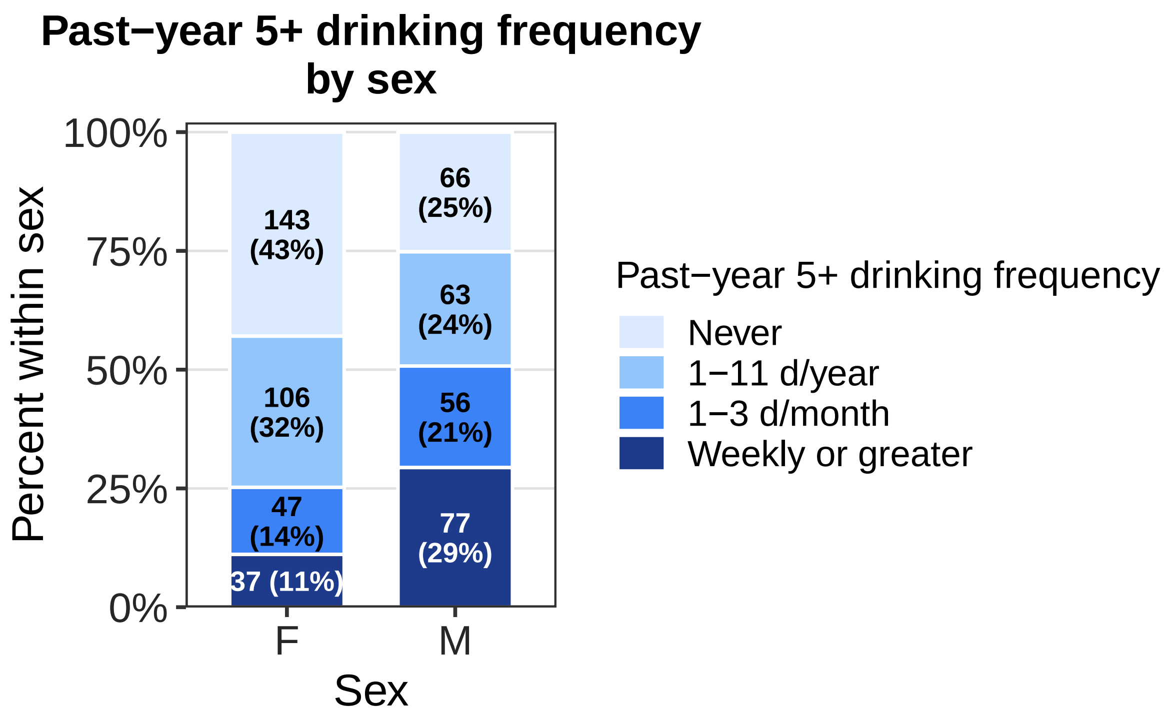

Figure 2: Distribution of past-year 5+ drinking frequency by sex.

Final Figures 3 And 4

This section displays the pooled and sex-stratified HPC-VS ordinal association figures with bootstrap confidence ribbons.

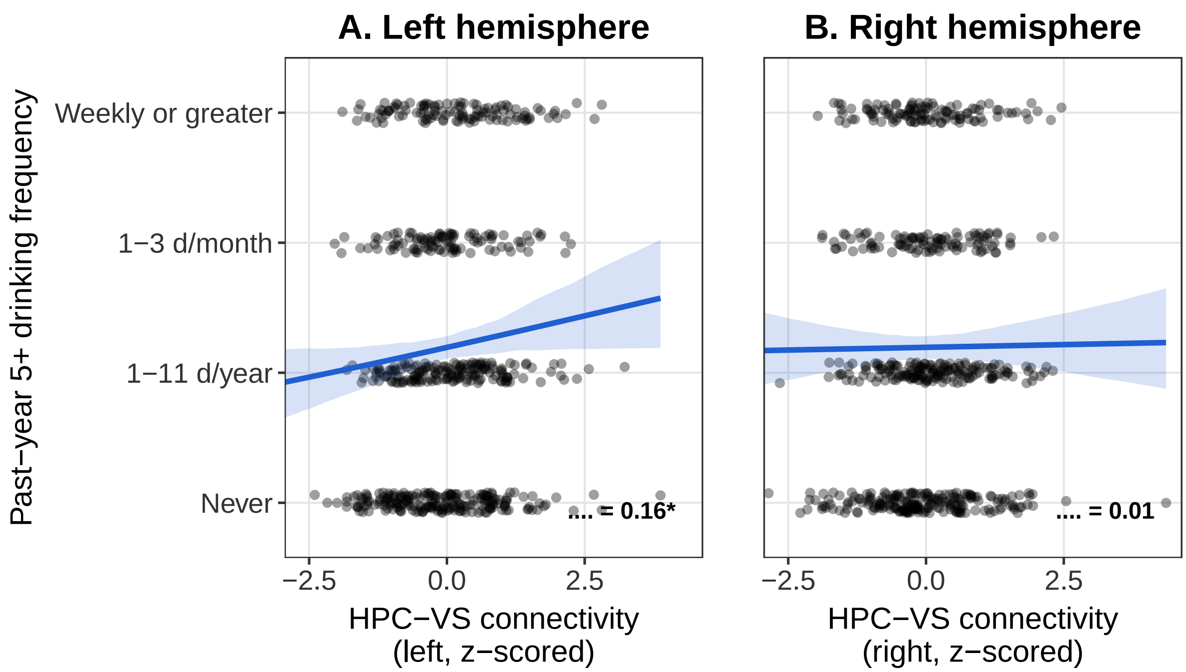

Figure 3: Pooled HPC-VS ordinal associations.

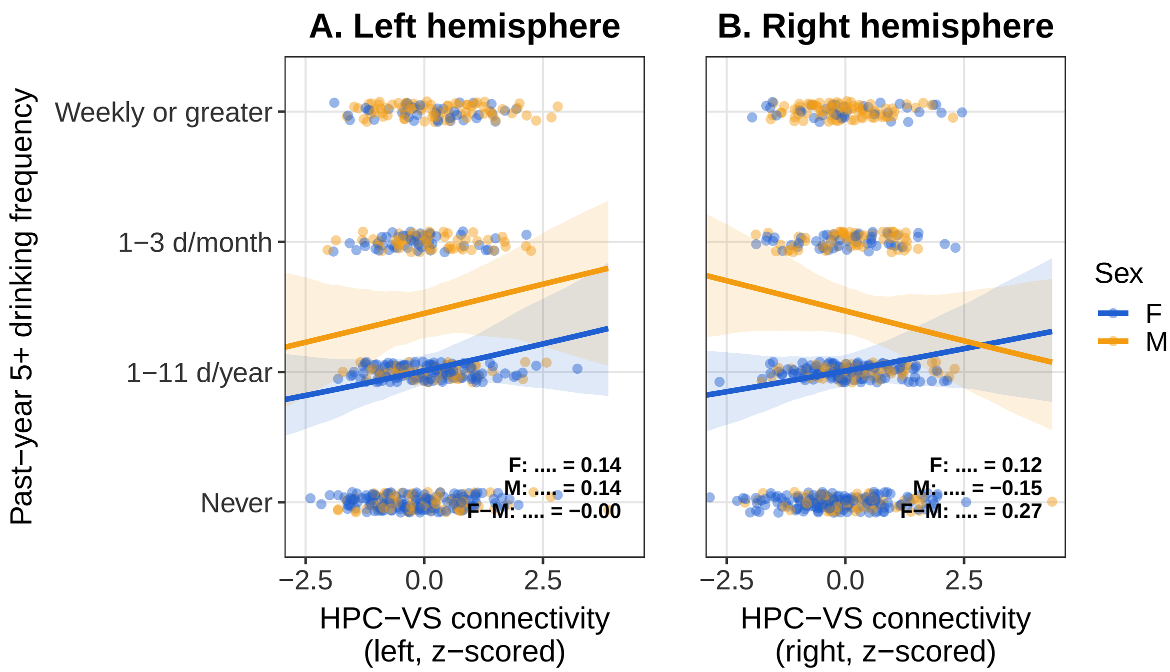

Figure 4: Sex-stratified HPC-VS ordinal associations.

Final Figure 5

This section creates the diagnosis-group HPC-VS connectivity figure reported in the manuscript.

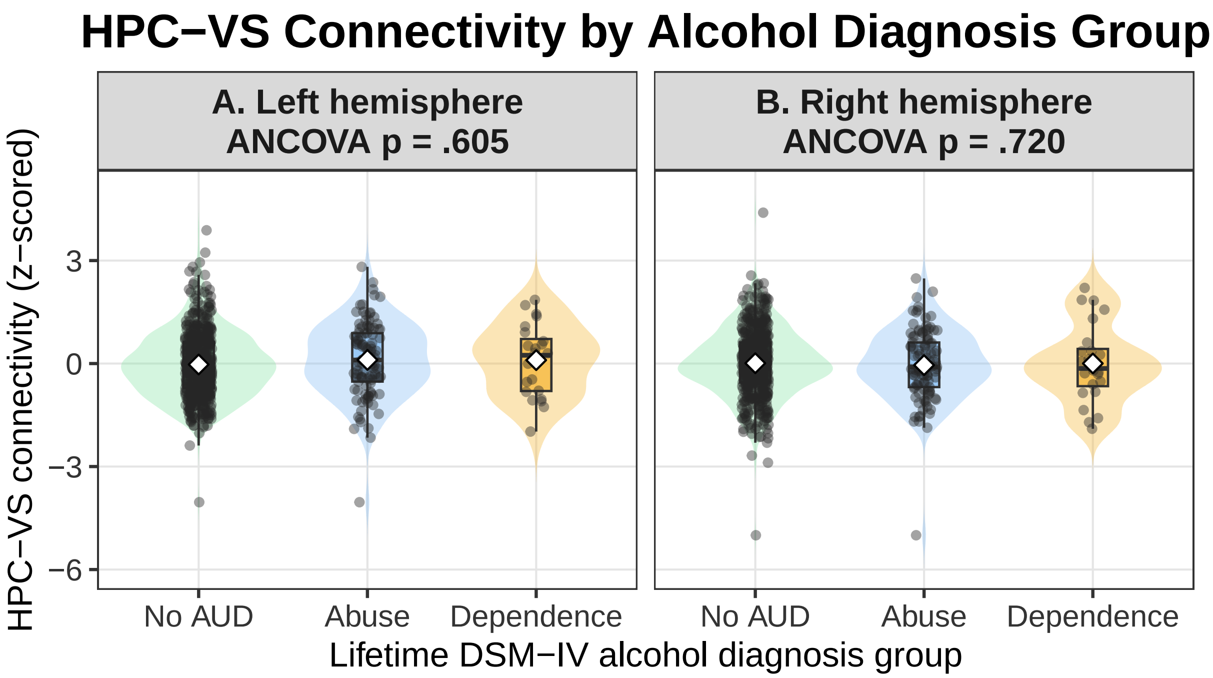

Figure 5: HPC-VS connectivity by alcohol diagnosis group.